# ENTERPRISE INFRASTRUCTURE PLAN

## Nexora Digital Solutions, Inc.

*IT Infrastructure Planning for a Newly Established Startup Company*

Prepared for: Company Management / IT Manager
Prepared by: Ryan A. Yasoña
Role: Junior System Administrator

ITEP 414 - System Administration and Maintenance
Week 2 Portfolio Project | Bachelor of Science in Information Technology
Laguna State Polytechnic University

# Part 1:
# Company Name: Nexora Digital Solutions, Inc.

**Nature of Business:**

Nexora Digital Solutions is a Philippine-based technology startup that develops web-based business management systems, cloud-enabled applications, and digital solutions for small and medium-sized businesses. The company specializes in developing customized systems for areas such as e-commerce, property management, finance, and internal business operations

**Vision**

> To become a trusted technology partner for businesses by delivering secure, accessible, and innovative digital solutions that simplify operations and support sustainable growth.

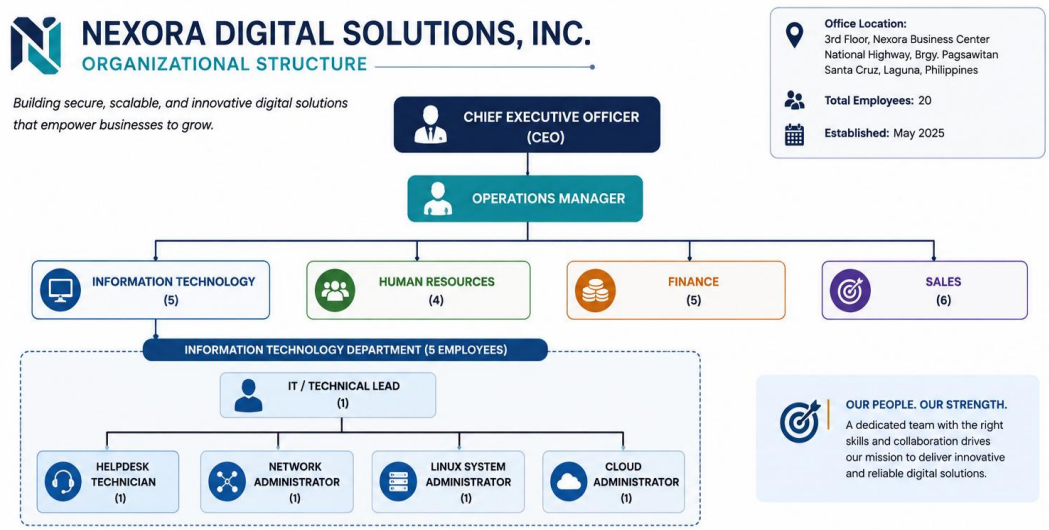
**Mission**

> Nexora Digital Solutions is committed to developing reliable and user-centered software solutions that help businesses improve their daily operations. Through modern web technologies, cloud services, secure IT infrastructure, and continuous innovation, the company aims to transform complex business processes into efficient digital experiences.

**Fictional Office Location**

**Nexora Digital Solutions, Inc.**
3rd Floor, Nexora Business Center
National Highway, Brgy. Pagsawitan
Santa Cruz, Laguna, Philippines

Organizational Structure



**Employee Distribution**

| Department | Number of Employees | Positions |
|---|---|---|
| Information Technology (IT) | 5 | IT/Technical Lead, Helpdesk Technician, Network Administrator, Linux System Administrator, Cloud Administrator |
| Human Resources (HR) | 4 | HR Manager, Recruitment Specialist, HR Assistant, Training & Employee Relations Officer |
| Finance | 5 | Finance Manager, Accounting Staff, Payroll Officer, Finance Assistant, Accounting Assistant |
| Sales | 6 | Sales Manager, 3 Sales Representatives, Business Development Officer, Sales/Admin Assistant |
| Total | 20 | — |

# Part 2:
# Enterprise Hardware Inventory

The proposed hardware inventory for Nexora Digital Solutions, Inc. is designed to support the daily operations of its 20 employees across the Information Technology, Human Resources, Finance, and Sales departments. The selection considers employee productivity, network reliability, data protection, business continuity, and future expansion.

| Asset ID | Hardware | Quantity | Department | Purpose |
|---|---|---|---|---|
| HW-001–015 | Desktop Computers | 15 | IT, HR, Finance, Sales | Primary workstation for employees performing development, administrative, financial, and sales-related tasks |
| HW-016–020 | Laptops | 5 | IT, Sales | Portable workstations for IT personnel and sales employees who may work in meetings, client visits, or other locations |
| SRV-001 | Server | 1 | IT | Hosts internal services such as file sharing, user management, application services, monitoring, and other essential infrastructure |
| NET-001 | Enterprise Router | 1 | IT | Connects the company's internal network to the Internet and manages routing, firewall, and network traffic |
| NET-002–003 | Managed Network Switches | 2 | IT | Provides wired network connectivity for computers, server, printers, NAS, and other network devices |
| PRN-001–003 | Network Printers | 3 | HR, Finance, Sales | Provides shared printing and scanning services for documents, financial records, HR documents, and sales materials |
| UPS-001–008 | Uninterruptible Power Supply (UPS) | 8 | IT, Finance | Protects critical IT equipment and selected workstations from power interruptions, surges, and sudden shutdowns |
| WAP-001–003 | Wireless Access Points | 3 | IT / Office-wide | Provides secure wireless connectivity throughout the office and supports employee laptops and authorized mobile devices |
| NAS-001 | Network Attached Storage (NAS) | 1 | IT | Provides centralized file storage, shared folders, and local backup storage for important company data |
| BAK-001–002 | External Backup Drives | 2 | IT | Provides an additional offline backup copy of critical files and system data |
| MON- | Monitors | 20 | IT, HR, | Provides external displays for |

| Asset ID | Hardware | Quantity | Department | Purpose |
|---|---|---|---|---|
| 001–020 | | | Finance, Sales | employees, improving productivity and allowing laptops to function as full desktop workstations |

## Justification of Hardware Selection

| Hardware | Short Justification |
|---|---|
| **Desktop Computers** | Provide reliable and cost-effective workstations for office-based employees. |
| **Laptops** | Support IT and Sales employees who need mobility for meetings, troubleshooting, and off-site work. |
| **Server** | Hosts company applications, files, databases, and other internal services. |
| **Router** | Manages traffic between the company network and the Internet. |
| **Managed Switches** | Connect wired devices and support VLANs for better network management and security. |
| **Printers** | Provide shared printing and scanning for HR, Finance, and Sales. |
| **UPS** | Protects critical equipment from power interruptions and sudden shutdowns. |
| **Wireless Access Points** | Provides reliable Wi-Fi coverage throughout the office. |
| **NAS Storage** | Provides centralized file storage and local backups. |
| **External Backup Drives** | Provide an additional offline copy of important company data. |
| **Monitors** | Provide displays for all employees and improve workstation productivity. |
| **Network Rack** | Organizes and protects servers and networking equipment. |
| **IP Phones** | Support internal communication and client calls. |
| **Conference Display** | Used for meetings, presentations, and client discussions. |
| **CCTV Cameras/NVR** | Improve physical security and monitor important office areas. |

# Part 3:
# Enterprise Software Inventory

The software environment of Nexora Digital Solutions, Inc. is designed to support software development, office productivity, system administration, networking, cybersecurity, data management, and remote technical support. The company will use a combination of commercial, open-source, and free software while maintaining standardized versions across company devices.

| Software | Version | License | Purpose |
|---|---|---|---|
| **Windows 11 Pro** | 25H2 | Proprietary / Commercial | Primary operating system for employee desktops and laptops. It provides business-oriented features, device management capabilities, security features, and compatibility with common business applications. Windows 11 25H2 is a supported Windows release. |
| **Ubuntu Server** | 26.04 LTS | Open Source | Operating system for the company's server infrastructure. It can host web applications, databases, internal services, development environments, and other server workloads. Ubuntu LTS releases are designed for long-term support and enterprise use. |
| **Microsoft Office** | Office 2024 | Proprietary / Commercial | Provides Microsoft Word, Excel, PowerPoint, and other productivity applications needed for documentation, financial spreadsheets, reports, presentations, and business operations. Office 2024 is supported through October 2029. |
| **Visual Studio Code** | 1.113.1 | MIT License / Free | Primary code editor for software developers. It supports programming languages, extensions, debugging, Git integration, and web development workflows. |
| **Git** | 2.51.0 | GPL-2.0 | Distributed version-control system used by developers to track code changes, manage branches, collaborate on projects, and maintain software history. |
| **GitHub Desktop** | 3.5.0 | MIT License / Free | Provides a graphical interface for Git and simplifies repository management, commits, branches, synchronization, and collaboration for developers. GitHub Desktop also provides organization deployment options such as an MSI installer for Windows. |
| **VirtualBox** | 7.2.2 | GPLv3 / Extension Pack License | Allows IT personnel and developers to create and manage virtual machines. It can be used for testing operating systems, server configurations, and development environments without affecting production systems. |
| **Google Chrome** | 151.0.7922.137 | Proprietary / Freeware | Primary web browser for accessing web applications, cloud services, development documentation, business platforms, and online communication tools. |
| **Microsoft Defender** | 4.18.26070.1004 | Included with Windows | Provides built-in antivirus and endpoint protection against malware and other common security threats. It is used as the standard endpoint security solution for company Windows computers. |

| Software | Version | License | Purpose |
|---|---|---|---|
| **AnyDesk** | 9.6.0 | Proprietary / Commercial | Enables authorized IT personnel to remotely access and troubleshoot employee computers. Remote access should be restricted to approved IT staff and protected with appropriate authentication and access controls. |
| **7-Zip** | 26.02 | LGPL / Open Source | Used for compressing and extracting files, reducing file sizes, and handling common archive formats for software development and administrative tasks. |
| **Python** | 3.14.6 | PSF License / Open Source | Used by the development team for web applications, automation, scripting, system administration utilities, and backend development. |
| **PHP** | 8.3.32 | PHP License / Open Source | Supports development of web-based business applications and backend services. It is suitable for building scalable systems for clients and internal company operations. |
| **Laravel** | 12.0.0 | MIT License / Open Source | PHP framework used by developers to build structured, maintainable, and secure web applications efficiently. |
| **MySQL Community Server** | 8.4 LTS | GPL / Open Source | Provides relational database services for storing application data, customer information, transactions, user accounts, and other structured business information. |

## Software Justification

### Windows 11 Pro

Windows 11 Pro will be the standard operating system for company desktops and laptops. Standardizing the operating system simplifies administration, software deployment, troubleshooting, security configuration, and employee support. The Pro edition is appropriate for a business environment because it provides features intended for organizational use.

### Ubuntu Server

Ubuntu Server will serve as the primary server operating system. It is particularly appropriate for Nexora because the company develops web-based applications and requires a stable environment for hosting applications, databases, development services, and internal infrastructure. Ubuntu's LTS model is appropriate for a startup that wants predictable maintenance and long-term security support.

### Microsoft Office

Microsoft Office is required by every non-technical department for daily business operations. HR can use Word and Excel for employee documentation and records, Finance can use Excel for financial analysis and reports, and Sales can use Word and PowerPoint for proposals and presentations.

**Visual Studio Code**

VS Code is the primary development environment for the IT department. It supports multiple programming languages and extensions, making it suitable for the company's web application development activities. It can also integrate with Git and other development tools.

**Git and GitHub Desktop**

Git provides the underlying version-control system used by developers, while GitHub Desktop provides a graphical interface for employees who prefer a simpler workflow. Together, they allow the IT department to maintain organized source-code repositories, track changes, collaborate on projects, and recover previous versions of code.

**VirtualBox**

VirtualBox is important for system administration and software testing. IT personnel can create isolated virtual machines for testing Linux configurations, server services, networking scenarios, and software before deploying changes to production systems.

**Google Chrome**

Chrome will be the standard web browser because much of the company's work will involve web-based applications, cloud services, documentation, client systems, and online communication platforms. Standardizing the browser also makes troubleshooting easier for IT personnel.

**Microsoft Defender**

Microsoft Defender provides endpoint protection for Windows computers. Using the built-in security solution helps establish a baseline security configuration across employee workstations without requiring a separate antivirus product for every device.

**AnyDesk**

AnyDesk allows authorized IT personnel to remotely assist employees when an issue cannot be resolved through normal instructions. This reduces response time and avoids unnecessary physical visits to individual workstations. Remote access must be controlled through company policies and limited to authorized personnel.

**7-Zip**

7-Zip provides a lightweight tool for compressing and extracting files. Developers can use it when handling project archives, while administrative employees can use it when exchanging compressed documents and files.

**Python**

Python supports automation, scripting, backend development, and system administration tasks. IT employees can use Python to automate repetitive administrative activities and create internal utilities.

**PHP and Laravel**

Because Nexora is a software development company specializing in web-based business solutions, PHP and Laravel are included in the standard development environment. Laravel provides a structured framework for developing maintainable web applications, while PHP provides the underlying programming platform.

**MySQL Community Server**

MySQL provides relational database services for the company's web applications. It can store structured information such as users, products, transactions, customer records, and application data. It also gives the development team a realistic database environment for developing and testing business systems.

### 3.4 Software Licensing Strategy

Nexora Digital Solutions, Inc. will follow a **mixed software licensing strategy**. Commercial software such as Windows 11 Pro and Microsoft Office will be properly licensed for business use, while open-source software such as Ubuntu Server, Git, Python, PHP, Laravel, and MySQL will be used according to their respective licenses.

The IT department will maintain a software inventory and license record to prevent unauthorized installations, expired licenses, and unapproved applications. All software installed on company-owned devices must be approved by the IT/Technical Lead or designated system administrator.

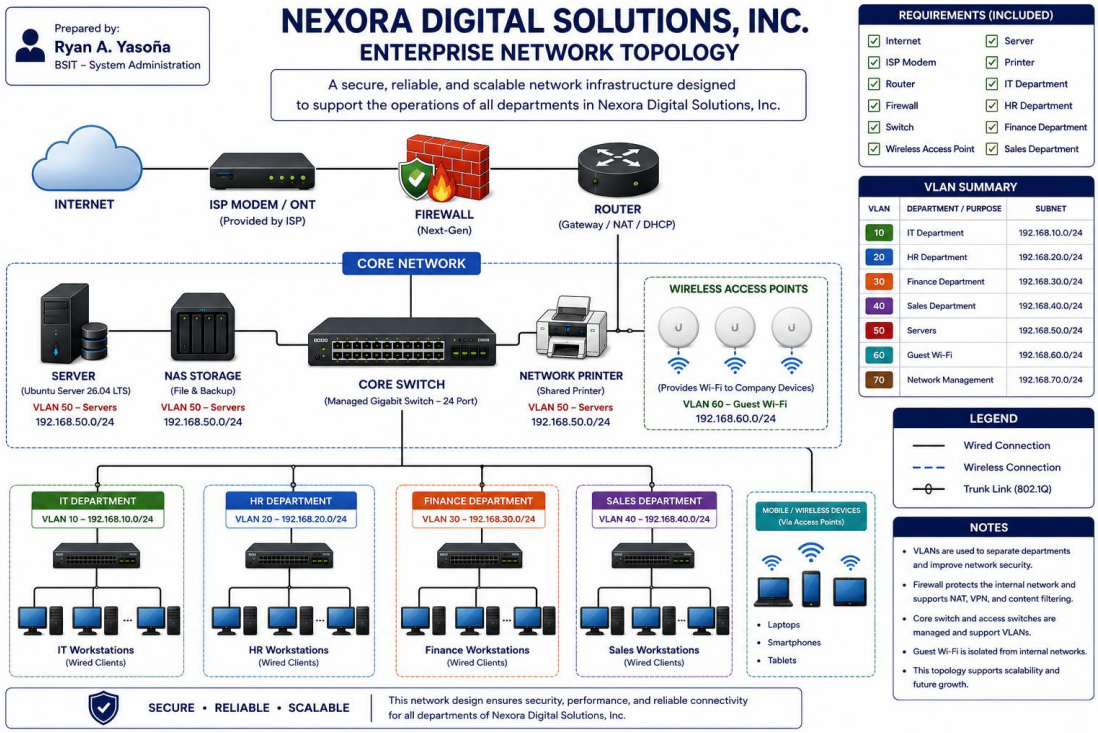
# Part 4:
# ENTERPRISE NETWORK INVENTORY

The proposed network infrastructure for Nexora Digital Solutions, Inc. is designed for a 20-employee single-floor office. The network will provide reliable wired and wireless connectivity while separating business functions through network segmentation. The design also provides sufficient capacity for future employees, additional devices, and expansion of the company's IT infrastructure.

| Asset ID | Network Equipment | Quantity | Department / Location | Purpose |
|---|---|---|---|---|
| NET-001 | ISP Modem / ONT | 1 | Network / Server Room | Receives the Internet connection from the Internet Service Provider and provides the connection to the company's network infrastructure |
| NET-002 | Business-Grade Router | 1 | Network / Server Room | Routes traffic between the internal network and the Internet, provides DHCP, NAT, VPN capabilities, and traffic management |
| NET-003 | Enterprise Firewall | 1 | Network / Server Room | Protects the internal network from unauthorized external access and filters incoming and outgoing network traffic |
| NET-004–005 | Managed Gigabit Switch | 2 | Network / Server Room | Provides wired connectivity for computers, servers, printers, access points, and other network devices; supports VLAN configuration |
| NET-006–008 | Wireless Access Point | 3 | Office Floor | Provides secure Wi-Fi coverage throughout the office for laptops and authorized wireless devices |
| NET-009–010 | 24-Port Patch Panel | 2 | Network / Server Room | Organizes and terminates Ethernet cables between office network outlets and the network switches |
| NET-011 | CAT6 Ethernet Cable | 1,000 m | Office / Network Cabling | Provides high-speed wired network connections between workstations, network outlets, patch panels, switches, and other devices |
| NET-012 | RJ45 Connectors | 200 pcs | IT / Network Storage | Used to terminate CAT6 Ethernet cables and create network patch cables |
| NET-013 | 24-Port Network Rack | 1 | Network / Server Room | Houses and organizes the switches, patch panels, router, firewall, and related networking equipment |
| NET-014 | Cable Management Panels | 2 | Network / Server Room | Keeps network cables organized and improves maintenance and troubleshooting |
| NET-015 | Gigabit Ethernet Patch Cables | 50 pcs | Office / Network Storage | Connects network devices and workstation cables to patch panels and switches |
| NET-016 | Network Cable Tester | 2 | IT Department | Tests Ethernet cables for continuity, wiring errors, and connection problems |
| NET-017 | Fiber/ISP Backup Connection | 1 | Network / Server Room | Provides an alternative Internet connection for business continuity if the primary ISP becomes unavailable |

# Part 5:
# Enterprise Network Diagram

The diagram below illustrates the logical enterprise network topology of **Nexora Digital Solutions, Inc.**, showing the connection from the Internet and ISP modem through the firewall, router, and core network before reaching the company's servers, wireless access points, printer, and departmental LAN segments. The topology was designed to provide a secure, reliable, and scalable network infrastructure for the IT, Human Resources, Finance, and Sales departments.




This network topology provides a structured and secure foundation for Nexora Digital Solutions, Inc.'s IT infrastructure. Through proper network segmentation, centralized management, wireless connectivity, and dedicated security controls, the proposed design supports the company's current operations while allowing room for future growth and expansion.

# PART 6:
# SYSTEM ADMINISTRATION ROLES

### Helpdesk Technician

**Responsibilities**

- Provide technical support to employees.
- Troubleshoot hardware, software, and connectivity problems.
- Install and configure workstations and applications.
- Create, update, and resolve IT support tickets.
- Assist users with account and password-related issues.
- Escalate complex technical problems to system or network administrators.

**Skills**

- Hardware and software troubleshooting
- Basic networking knowledge
- Windows operating system administration
- Customer service and communication
- Problem-solving
- Documentation and ticket management

**Common Tools**

- **AnyDesk** — remote technical support
- **Windows 11 Pro** — workstation administration
- **Microsoft Defender** — basic security troubleshooting
- **7-Zip** — file management
- **Ticketing systems** — tracking technical issues
- **Network diagnostic tools** — troubleshooting connectivity

**Certifications**

- **CompTIA A+**
- **Google IT Support Professional Certificate**
- **ITIL Foundation**
- **Microsoft 365 Certified: Fundamentals**

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### Network Administrator

**Responsibilities**

- Design, configure, and maintain the company's network.
- Configure routers, switches, firewalls, and access points.
- Manage VLANs, IP addressing, and network services.
- Monitor network performance and availability.

- Troubleshoot network connectivity problems.
- Implement network security controls.
- Maintain network documentation and topology diagrams.

**Skills**

- TCP/IP and networking fundamentals
- VLAN and subnetting
- Routing and switching
- Firewall configuration
- Wireless networking
- Network monitoring
- Network security and troubleshooting

**Common Tools**

- **Cisco IOS / network device CLI** — network configuration
- **Wireshark** — packet analysis
- **Ping / Traceroute** — connectivity testing
- **Nmap** — network discovery and security assessment
- **Network monitoring systems** — performance monitoring
- **SSH** — secure remote administration

**Certifications**

- **Cisco Certified Network Associate (CCNA)**
- **CompTIA Network+**
- **Juniper Networks Certified Associate (JNCIA)**
- **CompTIA Security+**

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**Linux System Administrator**

**Responsibilities**

- Install, configure, and maintain Linux servers.
- Manage users, groups, permissions, and system resources.
- Install and maintain server applications and services.
- Monitor server performance and availability.
- Perform system updates, backups, and recovery procedures.
- Troubleshoot Linux operating system and application issues.
- Maintain server security and access controls.

**Skills**

- Linux command-line administration
- Bash scripting
- File and permission management
- User and group administration

- Server configuration
- System monitoring
- Backup and recovery
- Basic networking and security

**Common Tools**

- **Ubuntu Server** — server operating system
- **Bash** — automation and administration
- **SSH** — remote server management
- **systemd** — service management
- **Cron** — scheduled tasks
- **Git** — configuration and code management
- **VirtualBox** — testing and virtualization

**Certifications**

- **Linux Professional Institute LPIC-1**
- **CompTIA Linux+**
- **Red Hat Certified System Administrator (RHCSA)**
- **Linux Foundation Certified System Administrator (LFCS)**

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**Cloud Administrator**

**Responsibilities**

- Manage the organization's cloud infrastructure and services.
- Configure cloud storage, virtual machines, and networking.
- Manage cloud users, permissions, and access controls.
- Monitor cloud resources and performance.
- Implement cloud backup and disaster recovery solutions.
- Control cloud resource usage and costs.
- Maintain cloud security and compliance.

**Skills**

- Cloud computing fundamentals
- Virtualization
- Cloud networking
- Identity and access management
- Cloud storage and backup
- Security and monitoring
- Automation
- Cost management

**Common Tools**

- **AWS Management Console** — cloud resource management
- **AWS CLI** — command-line cloud administration
- **Microsoft Azure Portal** — Azure resource management
- **AWS CloudWatch** — monitoring
- **Terraform** — infrastructure automation
- **Docker** — application containerization
- **Git** — infrastructure and configuration management

**Certifications**

- **AWS Certified Cloud Practitioner**
- **AWS Certified Solutions Architect – Associate**
- **Microsoft Certified: Azure Administrator Associate**
- **CompTIA Cloud+**

**Collaboration of the Four Roles**

The Helpdesk Technician, Network Administrator, Linux System Administrator, and Cloud Administrator work together to maintain a reliable IT environment. The Helpdesk Technician serves as the first point of contact for employees and handles common technical problems before escalating complex issues. The Network Administrator manages the company's network infrastructure and ensures that users and systems can communicate securely. The Linux System Administrator maintains the organization's Linux servers, applications, and internal services, while the Cloud Administrator manages cloud-based resources, storage, security, and services. By communicating and coordinating their activities, these four professionals can quickly identify problems, maintain system availability, protect company resources, and support the organization's daily operations.

# PART 7:
# INFRASTRUCTURE RECOMMENDATIONS

**Internet Provider**

Nexora Digital Solutions, Inc. should subscribe to a **business-grade fiber Internet service** with at least **500 Mbps to 1 Gbps** bandwidth and a Service Level Agreement (SLA).

**Recommendation:**

- Primary connection: Business Fiber — **500 Mbps minimum**
- Preferred capacity: **1 Gbps**
- Secondary connection: **100–300 Mbps backup Internet**
- Static public IP address for business services where required
- Business support and SLA

**Reason:**
As a software development company, Nexora relies heavily on cloud services, Git repositories, video meetings, software downloads, remote support, and online business applications. A business-grade connection provides better reliability and support than a typical residential connection. A secondary ISP can maintain basic operations if the primary connection fails.

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**Server Specifications**

Nexora should use a dedicated server capable of handling current workloads while allowing future expansion.

| Component | Recommended Specification |
|---|---|
| Processor | **Intel Xeon / AMD EPYC, 8+ cores** |
| RAM | **32 GB ECC RAM**, expandable to 64 GB |
| Storage | **4 × 1 TB Enterprise SSD, RAID 10** |
| Network | **2 × 1 Gbps Ethernet ports** |
| Power | Redundant power supply + UPS |
| Operating System | **Ubuntu Server 26.04 LTS** |
| Backup | Automated backup to NAS + external storage |

**Reason:**
The server will support internal applications, databases, file services, development environments, and other infrastructure services. ECC memory, RAID storage, and redundant power improve reliability and reduce the risk of downtime or data loss.

**Backup Strategy**

Nexora should implement the **3-2-1 backup strategy**:

> **3 copies of data → 2 different storage types → 1 copy stored off-site**

**Recommended Backup Plan**

| Backup | Frequency | Location |
|---|---|---|
| Server → NAS | Daily | On-site |
| NAS → External Drive | Daily/Weekly | On-site but disconnected when not backing up |
| Critical Data → Cloud/Off-site Storage | Weekly | Off-site |
| Full System Backup | Weekly | NAS / Cloud |
| Backup Restoration Test | Monthly | IT Department |

**Reason:**
Having multiple backup copies protects the company from hardware failure, accidental deletion, malware, ransomware, and other incidents. An off-site copy ensures that data can still be recovered if the office experiences a major incident.

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**Security Recommendations**

Nexora should implement a layered security approach rather than relying on a single security product.

Recommended controls include:

- Enterprise firewall
- VLAN network segmentation
- Secure Wi-Fi using WPA3 where supported
- Multi-factor authentication (MFA)
- Regular operating system and software updates
- Least-privilege access
- Regular vulnerability scanning
- Encrypted backups
- Secure remote access through VPN
- Restricted administrator accounts
- Employee security awareness training
- Physical security for the server/network room
- Regular security log monitoring

**Reason:**
Nexora handles source code, employee information, financial records, and client data. Layered security reduces the possibility that one compromised device or account can expose the entire company network.

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**Antivirus / Endpoint Protection**

The company should use Microsoft Defender Antivirus as the standard endpoint protection for Windows 11 Pro computers.

For business-wide management, Nexora should consider Microsoft Defender for Business or another centrally managed endpoint security solution as the company grows.

**Reason:**
Centralized endpoint protection allows IT administrators to monitor security status, manage policies, detect malware, and respond to threats across company devices. Microsoft Defender also integrates well with Windows 11 Pro.

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**Password Policy**

Nexora should establish a company-wide password policy.

**Recommended Requirements**

- Minimum **12 characters** for standard passwords
- Use a combination of letters, numbers, and special characters where appropriate
- Do not reuse company passwords across different services
- Enable MFA for email, cloud services, administrative accounts, and other critical systems
- Use a company-approved password manager
- Never share passwords with other employees
- Immediately change compromised passwords
- Disable accounts immediately when an employee leaves the company
- Administrative accounts must use separate credentials from normal user accounts

**Reason:**
Strong authentication reduces the risk of unauthorized access caused by stolen or easily guessed passwords. MFA provides an additional layer of protection even if a password is compromised.

**Expansion Plan**

Nexora should design its infrastructure so it can support growth from 20 employees to approximately 50 employees without requiring a complete network replacement.

**Short-Term Expansion — 20–30 Employees**

- Add additional workstations and laptops
- Add network switch capacity
- Expand NAS storage
- Increase Internet bandwidth if necessary
- Add additional wireless access points
- Increase server RAM and storage when required

**Medium-Term Expansion — 30–50 Employees**

- Add additional managed switches
- Expand server resources or deploy additional servers
- Implement centralized identity and device management
- Increase cloud infrastructure
- Deploy stronger endpoint security
- Consider a dedicated backup server
- Upgrade to higher-capacity Internet connections

**Long-Term Expansion — 50+ Employees**

- Consider redundant servers and network equipment
- Implement high-availability services
- Expand cloud infrastructure
- Introduce centralized Security Information and Event Management (SIEM)
- Establish a formal disaster recovery site or cloud-based disaster recovery solution

**Reason:**
Planning for growth prevents Nexora from repeatedly replacing its infrastructure as the company expands. Modular and scalable equipment allows the company to add users and services while minimizing disruption and unnecessary costs.

# PART 8:
# PERSONAL REFLECTION

This Week 2 portfolio project helped me understand that system administration is not only about fixing computers, configuring servers, or troubleshooting network problems. It also involves careful planning before any technology is deployed. Creating an IT infrastructure for Nexora Digital Solutions, Inc. from the beginning allowed me to experience how a System Administrator must consider the company's employees, departments, business operations, security needs, available resources, and future growth when making technical decisions.

One of the most important things I learned from this project was how different parts of an IT infrastructure are connected. The hardware inventory, software inventory, network equipment, and network topology cannot be planned independently. For example, determining the number of employees and their responsibilities helped me decide which devices each department would need. The network inventory then had to accommodate those devices, servers, printers, wireless access points, and other equipment. I also learned that network segmentation through VLANs can help improve security by separating departments and controlling access between them.

The most challenging task for me was creating the enterprise network diagram. It was more difficult than simply listing the networking equipment because I had to determine how each component should be connected and how information would flow through the network. I also had to make sure that the diagram was understandable, properly labeled, and included the required networking devices and departments. This helped me better understand the relationship between the physical and logical aspects of a network.

This project also showed me why planning is important before deployment. Proper planning prevents unnecessary expenses, configuration problems, security weaknesses, and future limitations. A System Administrator needs to think ahead instead of simply purchasing equipment based on immediate needs. Planning also makes it easier to explain and justify technical decisions to management.

Overall, this project improved my understanding of the responsibilities involved in system administration. I learned how to analyze business requirements, select appropriate hardware and software, design a network, consider security and backup strategies, and document technical decisions in a professional way. It also helped me develop a more systematic approach to solving IT problems. In the future, I believe these skills will help me become a more capable System Administrator because I will be able to plan infrastructure with both technical requirements and business objectives in mind.